Shubham Verma

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EDUCATION

Vellore Institute of Technology Bhopal, Madhya Pradesh
Bachelor of Technology in Computer Science Engineering *with Specialization in Health Informatics 4 Years Program* (2026)
Overall GPA: 7.92/10.0

SKILLS

Languages	Python, C++, Java, Kotlin, SQL, HTML/CSS
Python Frameworks and Libraries	BeautifulSoup , Seaborn , LangChain, Pandas, NumPy, OpenCV, Scikit-Learn, Flask, Pinecone, Pytorch,
Tools & Platforms	Git, GitHub, Google Colab, Android Studio, MySQL, HuggingFace

PROJECTS

- **Diabetes Risk Classifier (Custom KNN Implementation) | *Python, NumPy, Pandas*** ***Link***
  - Architected a custom K-Nearest Neighbors (KNN) classifier from scratch using NumPy, focusing on optimized distance-vector calculations for medical diagnostic data.
  - Achieved a predictive accuracy of 69.83 through hyperparameter tuning (k=5) and rigorous validation on standardized clinical features (Glucose, BMI, Age).
  - Conducted Exploratory Data Analysis (EDA) to address class imbalances and visualize feature correlations, ensuring robust data preprocessing for the classification engine.
- **Medical Chatbot (End-to-End RAG Pipeline) | *Python, LangChain, Pinecone, Google Gemini*** ***Link***
  - Architected a Retrieval-Augmented Generation (RAG) system tailored for the healthcare domain, mitigating LLM hallucinations by grounding responses in verified medical literature ("The Gale Encyclopedia of Medicine").
  - Engineered a vector search pipeline using LangChain for text chunking and Pinecone as the vector database, enabling sub-second semantic retrieval of relevant medical context.
  - Integrated Google Gemini Pro to synthesize retrieved data into accurate, context-aware answers, serving the model via a robust web interface.

EXTRACURRICULAR ACTIVITIES

- Member Of Data Science Club
- Data Science Nights 2.0 (25 November 2023)
- Participated in "Corporate Role Play" (16 November 2022)
- Participated in "ECO - Hackathon" (March 2025) certificate
- Participated in "L'Oreal Sustainability Challenge" (February 2025) certificate

EXPERIENCE

Android Development Intern | Ashva Wearable Technologies Pvt Ltd. ***Sep 2024 – Jan 2025***

- **Project: Hip Navigation**
  - Engineered a cross-platform Android application using Kotlin Multiplatform (KMP) to interface with two distinct firmware architectures.
  - Designed a responsive User Interface (UI) utilizing Jetpack Compose and XML for optimal user experience.

Software Development Intern | Neuranics Lab February 2026 - Present

- **Project: Deriving CBC Parameters Through Deep Learning**
  - Optimized inference speed and overall performance by migrating complex computational codebases from CPU to GPU architectures.
  - Implemented Computer Vision (CV) techniques to generate precise image masks, enabling accurate automated background correction.
  - Successfully deployed the accelerated vision pipeline onto an NVIDIA Jetson Orin Nano for efficient edge computing and real-time processing.

CERTIFICATIONS

- **Applied Machine learning in python (Coursera Course) certificate**
- **Introduction to C++ (Simplilearn Course)**
- **Python Essentials (Vityarthi Course)**
- **Data Science with Python (iamneo)**
- **Computer Vision(VITyarthi Course)**
- **Java Development on Oracle Cloud (Oracle) certificate**
- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate(Oracle) certificate**
- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional (Oracle) certificate**
- **Oracle Cloud Infrastructure 2025 Data Science Professional (Oracle) certificate**
- **Oracle AI Vector Search Professional (Oracle) certification**